ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** water, air and a suitable temperature	Conditions needed for life are present.
2	**(c)** nitrogen	About 78 per cent.
3	**(a)** a small part of the total water	About 3 per cent is fresh and little is accessible.
4	**(a)** crust	Thin solid layer.
5	**(a)** 21 per cent	One-fifth.
6	**(a)** ice caps and glaciers	Frozen.
7	**(c)** mantle	Thickest layer.
8	**(c)** troposphere	Where we live.
9	**(c)** groundwater	Stored underground.
10	**(c)** inner core	Solid, very hot.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Crust (1); Mantle and core (1).	1+1
12	It supports life (1); It has water, air and suitable temperature (1).	1+1
13	Provides oxygen for breathing (1); Protects from harmful rays and temperature extremes (1).	1+1
14	Envelope of gases around the Earth (1); Nitrogen and oxygen (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Crust: thin and solid (1); Mantle: thick, hot, semi-molten rock (1); Core: outer liquid, inner solid, iron and nickel (1).	1+1+1
16	Not too hot or too cold (1); Water remains liquid (1); Habitable zone (1).	1+1+1
17	Earth has an atmosphere; the Moon has none (1); Earth has liquid water (1); Moon has extreme temperatures (1).	1+1+1
	OR: Nitrogen about 78 per cent (1); Oxygen about 21 per cent (1); Argon, carbon dioxide and other gases about 1 per cent (1).	1+1+1
18	Most water is in oceans and is salty (1); Most fresh water is frozen in ice caps (1); Little is in rivers and lakes (1).	1+1+1
	OR: Deflects charged particles from the Sun (1); Protects the atmosphere (1); Helps compass and migrating animals (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Crust: solid, thin (1); Mantle: thick, hot rock (1); Outer core: liquid iron and nickel (1); Inner core: solid (1); Studied through earthquake waves (1).	1+1+1+1+1
	OR: Absorbs UV (1); CFCs destroy ozone (1); Ozone hole over Antarctica (1); Montreal Protocol banned CFCs (1); Recovery is underway (1).	1+1+1+1+1
20	Troposphere: weather, life (1); Stratosphere: ozone layer (1); Mesosphere: meteors burn (1); Thermosphere: very hot, auroras (1); Exosphere: merges with space (1).	1+1+1+1+1
	OR: Deforestation (1); Burning fuels (1); Mining (1); Pollution (1); Overuse of water (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** stratosphere	Lies above the troposphere.
2	**(c)** condensation	Clouds.
3	**(a)** liquid	Molten metals.
4	**(c)** harmful ultraviolet rays of the Sun	Absorbs UV.
5	**(a)** harmful charged particles of the solar wind	Acts as a shield.
6	**(b)** iron and nickel	Dense metals.
7	**(b)** decreases	Air gets colder.
8	**(c)** liquid	Habitable zone.
9	**(a)** atmosphere	Air.
10	**(a)** mesosphere	Friction heats them.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	All water on the Earth (1); Oceans, rivers, lakes, groundwater, ice (1).	1+1
12	Rivers (1); Groundwater or glaciers (1).	1+1
13	Absorbs harmful ultraviolet rays (1); Protects living organisms (1).	1+1
14	The rocky outer layer of the Earth (1); Crust and uppermost mantle (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	The Earth's surface heats air from below (1); Farther from the ground there is less heat (1); Air is thinner higher up (1).	1+1+1
16	Biosphere: zone of living organisms (1); Hydrosphere: all water bodies (1); Biosphere includes parts of all spheres (1).	1+1+1
17	Smog (1); Acid rain (1); Breathing diseases (1).	1+1+1
	OR: Absorbs UV rays (1); CFCs from refrigerators and aerosols damage it (1); Leads to skin cancer and crop damage (1).	1+1+1
18	Rainwater harvesting (1); Stopping leakages and wastage (1); Reusing water (1).	1+1+1
	OR: Water cycle brings rain (1); Rivers carry water (1); Oceans moderate climate (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Lithosphere: rocks (1); Hydrosphere: water (1); Atmosphere: air (1); Biosphere: life (1); Interactions: rain and soil, plants and air (1).	1+1+1+1+1
	OR: Oxygen for breathing (1); CO_2 for plants (1); Ozone shield (1); Moderates temperature (1); Water vapour and weather (1).	1+1+1+1+1
20	Earth: liquid water, oxygen, moderate temperature (1); Venus: too hot because of CO_2 (1); Mars: cold and thin atmosphere (1); Distance from the Sun (1); Magnetic field and atmosphere (1).	1+1+1+1+1
	OR: About 4.5 billion years ago (1); Hot molten Earth cooled (1); Dense materials sank to form the core (1); Gases formed the atmosphere (1); Water vapour condensed to form oceans (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** very cold and dry	No liquid water.
2	**(d)** biosphere	Life zone.
3	**(d)** lower stratosphere or upper troposphere	Calm air.
4	**(a)** its thick carbon dioxide atmosphere	Strong greenhouse effect.
5	**(b)** about 71 per cent	Mostly oceans.
6	**(c)** greenhouse gases	e.g. CO2.
7	**(a)** day, night and seasons	Motions.
8	**(a)** lithosphere	Crust and upper mantle.
9	**(c)** carbon dioxide	CO2.
10	**(c)** 4.5 billion years	Formed long ago.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It rotates on its axis (1); Half faces the Sun at a time (1).	1+1
12	Troposphere, stratosphere (1); Mesosphere, thermosphere, exosphere (1).	1+1
13	Trapping of heat by gases in the atmosphere (1); Keeps the Earth warm (1).	1+1
14	Melting of glaciers (1); Rise in sea level and extreme weather (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Gases trap heat radiated by the Earth (1); Carbon dioxide, methane, water vapour (1); Keeps the Earth warm enough for life (1).	1+1+1
16	Burning fossil fuels (1); Deforestation (1); Planting trees and using clean energy (1).	1+1+1
17	Using resources so that future generations also have them (1); Solar energy (1); Recycling (1).	1+1+1
	OR: Liquid water (1); Air with oxygen (1); Moderate temperature (1).	1+1+1
18	Ozone absorbs UV (1); Meteors burn due to friction in the atmosphere (1); Atmosphere keeps temperature moderate (1).	1+1+1
	OR: Troposphere decreases (1); Stratosphere increases due to ozone (1); Mesosphere decreases (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Greenhouse effect is natural (1); Excess gases cause global warming (1); Melting ice, rising sea (1); Changing climate patterns (1); Reduce emissions, plant trees (1).	1+1+1+1+1
	OR: Earth: 24 h day, 365 days (1); Moon: no atmosphere, long day (1); Mars: 24.6 h day, longer year (1); Seasons from tilt (1); Conditions differ (1).	1+1+1+1+1
20	71 per cent of the surface is water (1); Most is salt water (1); Fresh water is mostly frozen (1); Water cycle (1); Need for conservation (1).	1+1+1+1+1
	OR: Water: harvesting, reuse (1); Soil: terracing, mulching (1); Forests: afforestation, protection (1); Air: fewer fuels, clean energy (1); Public participation (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.